

Pablo Zúñiga Carvajal

Industrial & Systems Engineer, BSc & MSc Equivalent
Data Modeling · Applied Statistics · Complex Systems
Valparaíso, Chile · Open to Relocation · US ESTA & H-1B1 Eligible
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Tech Stack & Core Competencies

- **Languages & Applied Mathematics:** Python (FastAPI, Polars, SciPy), TypeScript, SQL, R (Tidyverse, targets, forecast), Inferential Modeling, Time-Series, MILP & Optimization, Discrete-Event & Monte Carlo Simulations
- **Enterprise Data Architecture:** Containerization (Docker), ETL/ELT Pipelines, Metadata & Ontology Modeling, Data Governance, Enterprise Lineage, Relational Databases, Spatial & Telemetry Data Formats (.gpx, .fit, .kmz)
- **Cloud & Orchestration:** MS Azure, Git/GitHub (Actions CI/CD, Pages), Environment Orchestration
- **Systems Architecture & Engineering:** Systems Theory & Dynamics, Process Engineering (BPMN), System Diagramming (UML), Knowledge Architectures, Technical Documentation (Quarto, Markdown, HTML/CSS)

Work Experience

Cuxhaven Labs | Principal Data & Systems Consultant | Remote **May 2026 – Present**

Architected production-grade ETL/ELT pipelines and metadata ontologies for critical infrastructure under strict auditability standards. Engineered operational resilience frameworks and disaster recovery protocols to guarantee continuous operational uptime during system incidents.

Independent Forward-Deployed Engineering Consultant | Remote **Jul 2022 – Present**

Telemetry Analytics Pipeline: Built an end-to-end time-series analytics pipeline in Python/R processing pace-distance telemetry across 1.8k+ athletes to quantify physiological gaps and optimize pacing strategies under multi-variable environmental conditions. Designed relational database structures, automated data flows, and technical documentation architectures using Git/Markdown.

Decentralized Systems & Protocols | On-Chain Operations Specialist | Remote **Jan 2020 – Jul 2022**

Managed multi-sig treasury architecture (\$50k+) across Polkadot, Ethereum, and BSC. Executed dynamic gas optimization, cross-chain messaging, and smart contract coordination for 1,100+ concurrent users.

Banco de Crédito e Inversiones | Process Engineer (Data Governance) | Santiago, Chile **May 2019 – Sep 2019**

Orchestrated automated metadata integration pipelines from core transactional systems into MS Azure, standardizing BPMN workflows. Audited data architecture compliance in regulated banking environments, enforcing schema consistency and RBAC security controls.

ESVAL (Water Utility) | Lead Data & Project Engineer | Valparaíso, Chile **Jul 2017 – Dec 2017**

Developed a spatio-temporal risk model using Cox Proportional Hazards in R across 5,300 km of pipeline serving 1.9M+ residents. Delivered predictive maintenance risk scores via Agile/PMO frameworks to prioritize capital expenditure allocation for regional grid resilience.

ENAMI & Antofagasta Minerals | Industrial Engineering Intern | Chile **2014 & 2016**

Audited metallurgical inventory database (75k+ SKUs), uncovering 2.5M USD in accounting discrepancies for corporate financial audits. Monitored open-pit mining geotechnical telemetry for slope failure prediction and early risk mitigation in mission-critical environments.

Education

Pontificia Universidad Católica de Valparaíso | Valparaíso, Chile

Industrial & Systems Engineering (Integrated 6-Year Program, BSc & MSc Equivalent) | 2011 – 2017

Core Academic Foundations: Statistics, Systems Theory, Operations Research, Econometrics, Systems Modeling.

Leadership & Teaching

Graduate Teaching Assistant (PUCV)

Instructed Applied Data Science, MIS, and Quantitative Mathematics for postgraduated students (2018–2021).

Technical Instructor (BancoEstado / UTFSM)

Delivered time-series analytics and high-frequency financial modeling training to institutional trading desk staff.